<u>Prop</u> Let $(F, +, \cdot)$ be a field. Then:

① The additive and multiplicative identities are unique.

② Additive and multiplicative inverses are unique.

③ If $a, b, c \in F$ satisfy $a+b = a+c$, then $b=c$.
If $a, b, c \in F$ satisfy $a \cdot b = a \cdot c$ and $a \neq 0$, then $b=c$.

④ $a \cdot 0 = 0 \cdot a = 0 \quad \forall a \in F$

⑤ $(-a) \cdot b = a \cdot (-b) = -(a \cdot b) \quad \forall a, b \in F$

⑥ $(-a) \cdot (-b) = a \cdot b \quad \forall a, b \in F$

⑦ If $a \cdot b = 0$ then $a = 0$ or $b = 0$.

<u>Pf</u>: ① Additive identity: Suppose $0, 0' \in F$ satisfy

$$a + 0 = 0 + a = a \quad \forall a \in F \quad \text{(i)}$$
$$a + 0' = 0' + a = a \quad \forall a \in F \quad \text{(ii)}$$

Take $a = 0'$ in (i): $0' + 0 = 0'$

Take $a = 0$ in (ii): $0' + 0 = 0$

Together, we conclude $0 = 0'$.

Multiplicative identity: HW2.

② Fix $a \in F$. Suppose $-a, b \in F$ satisfy

$$-a + a = a + (-a) = 0, \qquad b + a = a + b = 0.$$

Then

$$b + a = 0 \quad \overset{+(-a)}{\Longrightarrow} \quad (b+a) + (-a) = 0 + (-a)$$

$$\overset{A3,\ A4}{\Rightarrow} \quad b+(a+(-a)) = -a$$
$$\overset{A5}{\Rightarrow} \quad b+0 = -a$$
$$\overset{A4}{\Rightarrow} \quad b = -a$$

Multiplicative inverses: HW2.

(3) Suppose $a+b=a+c$. Want: $b=c$.

$$a+b = a+c \quad \overset{(-a)+}{\Rightarrow} \quad -a+(a+b) = -a+(a+c)$$
$$\overset{A3}{\Rightarrow} \quad (-a+a)+b = (-a+a)+c$$
$$\overset{A5}{\Rightarrow} \quad 0+b = 0+c$$
$$\overset{A4}{\Rightarrow} \quad b = c$$

The second statement: HW2.

(4) Let $a \in F$. Then

$$0\cdot a \overset{A4}{=} (0+0)\cdot a \overset{D}{=} 0\cdot a + 0\cdot a$$

So by (3) we have $0\cdot a = 0$. Note $a\cdot 0 = 0\cdot a$ by M2, so $a\cdot 0 = 0$ as well.

(5) Let $a,b \in F$. Then

$$(-a)\cdot b + a\cdot b \overset{D}{=} (-a+a)\cdot b \overset{A5}{=} 0\cdot b \overset{(4)}{=} 0$$

So $(-a)\cdot b = -(a\cdot b)$. Similarly, $a\cdot(-b) = -(a\cdot b)$.

(6) Let $a,b \in F$. Then

$$(-a)\cdot(-b) + (-(a\cdot b)) \overset{(5)}{=} (-a)\cdot(-b) + a\cdot(-b)$$
$$\overset{D}{=} (-a+a)\cdot(-b)$$
$$\overset{A5}{=} 0\cdot(-b)$$

$$\overset{④}{=} 0$$

So $(-a)\cdot(-b) = a\cdot b$.

⑦ Suppose $a\cdot b = 0$. Want: $a=0$ or $b=0$.

If $a=0$ then we're done, so assume $a\neq 0$.

Then $\exists a^{-1}\in F$ s.t. $a\cdot a^{-1} = a^{-1}\cdot a = 1$.

$$\begin{aligned}
a\cdot b = 0 &\overset{a^{-1}\cdot}{\Longrightarrow} a^{-1}\cdot(a\cdot b) = a^{-1}\cdot 0 \\
&\overset{M3,\ ④}{\Longrightarrow} (a^{-1}\cdot a)\cdot b = 0 \\
&\overset{M5}{\Longrightarrow} 1\cdot b = 0 \\
&\overset{M4}{\Longrightarrow} b = 0. \qquad \square
\end{aligned}$$

<u>Def</u> An <u>order relation</u> $<$ on a nonempty set A satisfies the following properties:

① trichotomy: if $a,b\in A$, then exactly one of the following statements holds: $a<b$ or $a=b$ or $b<a$.

② transitivity: if $a,b,c\in A$ s.t. $a<b$ and $b<c$, then $a<c$.

<u>Ex</u> For $a,b\in\mathbb{Z}$, we write

$$a<b \iff b-a\in\mathbb{N}.$$

Then $<$ is an order relation on $\mathbb{Z}$.

<u>Notation</u> We write $a>b$ if $b<a$

$a\leq b$ if $(a<b$ or $a=b)$

$a\geq b$ if $b\leq a$

Def Let $(F, +, \cdot)$ be a field. We say $(F, +, \cdot)$ is an <u>ordered field</u> if it is equipped with an order relation $<$ that satisfies:

(O1) if $a, b, c \in F$ s.t. $a < b$, then $a + c < b + c$

(O2) if $a, b, c \in F$ s.t. $a < b$ and $0 < c$, then $a \cdot c < b \cdot c$

Ex $\mathbb{Q}$ is an ordered field. We define

$$\frac{a}{b} > 0 \iff a \cdot b \in \mathbb{Z} \text{ satisfies } a \cdot b > 0.$$

We have to check that this is well-defined:

If $\frac{a}{b} = \frac{c}{d}$ and $\frac{a}{b} > 0$, then $\frac{c}{d} > 0$:

$$(a,b) \sim (c,d) \implies ad = bc$$

$$\implies 0 < (ad)^2 = abcd$$

Note $ad \neq 0$, since $\begin{cases} d \neq 0 \\ ab > 0 \implies a \neq 0 \end{cases}$

So $\left.\begin{array}{l} 0 < (ab) \cdot (cd) \\ 0 < ab \end{array}\right\} \implies cd > 0 \implies \frac{c}{d} > 0.$

Next, we define

$$\frac{a}{b} < \frac{c}{d} \iff \frac{c}{d} - \frac{a}{b} > 0.$$

Exc:
- This $<$ is an order relation on $\mathbb{Q}$.
- (O1) and (O2) hold.

Prop Let $(F, +, \cdot, <)$ be an ordered field. Then:

① $a > 0 \iff -a < 0$

② If $a, b, c \in F$ satisfy $a<b$ and $c<0$, then $a \cdot c > b \cdot c$.

③ If $a \in F \setminus \{0\}$ then $a^2 = a \cdot a > 0$.
In particular, $1 > 0$.

④ If $a, b \in F$ satisfy $0<a<b$, then $0<b^{-1}<a^{-1}$.

<u>Pf</u>: ① $\Rightarrow$|: Suppose $a>0$. Then

$$a>0 \overset{O1}{\Rightarrow} a+(-a) > 0+(-a) \overset{A5, A4}{\Rightarrow} 0 > -a.$$

$\Leftarrow$|: Suppose $-a<0$. Then

$$-a<0 \overset{O1}{\Rightarrow} -a+a < 0+a \overset{A5, A4}{\Rightarrow} 0<a.$$

② Next time!